

168-GBd PCS-324QAM Signal Transmission with 2.1-Tb/s Net Bitrate Using Sample-Domain Volterra Nonlinear Pre-distortion

Masanori Nakamura, Fukutaro Hamaoka, Takayuki Kobayashi, and Etsushi Yamazaki

Network Innovation Laboratories, NTT, Inc., Yokosuka, Japan, msnr.nakamura@ntt.com

Abstract We demonstrate 168-GBd PCS-324QAM signal generation and transmission using sample-domain Volterra nonlinear pre-distortion applied to the oversampled high-PAPR waveform. Our approach provides a 0.7-dB peak-SNR gain over a symbol-domain Volterra filter and achieves net bitrates of 2.18 Tb/s back-to-back and 2.10 Tb/s after 80-km transmission. ©2026 The Author(s)

Introduction

Digital coherent transmission with high symbol rates and high-order probabilistically constellation-shaped (PCS-)QAM formats is a key approach toward multi-terabit single-carrier channels for next-generation optical networks. Advances in high-speed electrical signalling and PCS-QAM have pushed the reported per-wavelength net bitrates beyond 2.5 Tb/s [1]. Figure 1 plots recently reported net bitrates as a function of the information rate, defined as the net bitrate per 4D symbol [1–11]. These demonstrations reflect progress along two complementary directions: increasing the symbol rate through electrical-domain bandwidth multiplexing [1, 9], and increasing the information rate per symbol through improvements in signal quality such as low- V_π and wideband thin-film lithium-niobate (TFLN) modulators [10] and advances in transmitter- and receiver-side digital signal processing [4, 11].

Increasing the information rate at a given symbol rate requires higher transmitter signal quality. Nonlinear distortions at the optical transmitter arise mainly from the driver amplifier operating close to saturation and the nonlinear transfer characteristic of the optical modulator. Volterra-series-based digital pre-distortion, widely used for RF power amplifiers [12], was adapted to optical transmitter components with kernels operating on oversampled signals [13]. More recently, experimental demonstrations at high symbol rates have performed compensation

at one sample per symbol, including Volterra-filter-based [4], neural-network-based [11], and look-up-table-based [14] approaches. However, the transmitter waveform after Nyquist pulse shaping and linear pre-equalization exhibits a high peak-to-average power ratio (PAPR) with a continuous amplitude distribution, which limits the compensation achievable with symbol-domain digital pre-distortion.

In this paper, we demonstrate 168-GBd PCS-324QAM signal generation and 80-km transmission using sample-domain Volterra nonlinear pre-distortion that directly operates on the high-PAPR sample sequence. Owing to the continuous-amplitude distribution of these samples, the sample-domain Volterra filter provides a 0.7-dB peak-SNR gain over a symbol-domain Volterra filter at 168-GBd uniform 64-QAM. We achieved net bitrates of 2.18 Tb/s at back-to-back and 2.10 Tb/s after 80-km single-span transmission (see Fig. 1).

Principle

Nonlinear distortions at an optical transmitter arise mainly from gain compression and memory effects in the driver amplifier and the nonlinear transfer characteristic of the optical modulator. While linear pre-equalization compensates the transmitter's linear frequency response, at high symbol rates the pre-equalized Nyquist-shaped waveform exhibits a high-PAPR, continuous-amplitude distribution with instantaneous peaks above the symbol levels, which aggravates the

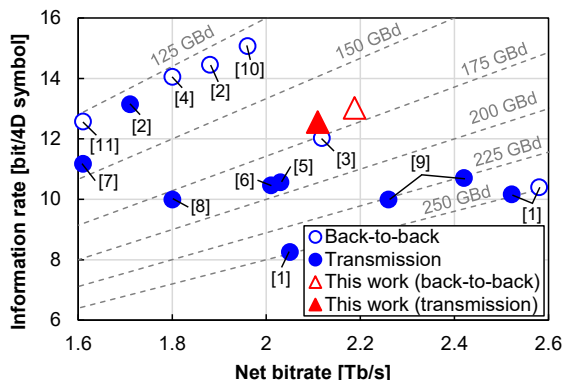


Fig. 1: Information rate (bit/4D symbol) vs. net bitrate per wavelength for recent high-symbol-rate experiments.

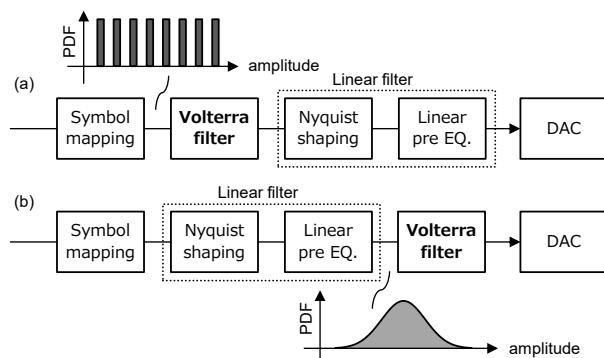


Fig. 2: Volterra filter configurations for Tx nonlinear pre-distortion: (a) symbol-domain VF, (b) sample-domain VF.

nonlinear distortion. Figure 2 compares two configurations of a Volterra filter for nonlinear digital pre-distortion: (a) a symbol-domain Volterra filter (symbol VF) placed before the Nyquist shaping filter, operating on the discrete-amplitude symbols; and (b) a sample-domain Volterra filter (sample VF) placed after the Nyquist shaping filter and linear pre-equalizer, directly operating on the continuous-amplitude sample sequence delivered to the digital-to-analogue converter (DAC). Because the waveform presented to the driver amplifier has a high-PAPR, continuous-amplitude distribution, the sample VF is expected to yield a larger SNR improvement than the symbol VF.

For the third-order sample-domain Volterra filter in this work, the output $y[n]$ is expressed as

$$y[n] = \sum_{p=1}^3 \sum_{k_1, \dots, k_p=0}^{m_p-1} h_p[k_1, \dots, k_p] \prod_{i=1}^p x[n - k_i]$$

where $x[n]$ and $y[n]$ denote the real-valued input and output digital sample sequences of each I and Q path, respectively, $h_p[\cdot]$ are the p -th-order Volterra kernels, and m_p are the corresponding memory lengths. Note that the first-order kernel $h_1[k]$ is a linear filter operating in series with the upstream linear pre-equalization. The pre-equalizer handles the bulk of the transmitter's linear frequency response, while $h_1[k]$ absorbs any residual linear component. The kernel coefficients are estimated via an indirect learning architecture (ILA) with the pre-equalized waveform as the reference [12, 13]. This preserves the high-PAPR, continuous-amplitude structure at the modulator input.

Experimental setup

Figure 3(a) shows the experimental setup for 168-GBd PCS-32QAM signal generation and transmission. A continuous-wave light with an optical frequency of 193.65 THz and a linewidth of ~ 10 kHz was generated from a micro-integrable tunable laser assembly (μ ITLA) and amplified to 19 dBm using a polarization-maintaining erbium-doped fibre amplifier (PM-EDFA). The amplified light drove a thin-film lithium-niobate (LN) in-phase and quadrature modulator (TFLN-IQM) with a 1-mm connector, showing an electrical-optical-electrical (EOE) response roll-off of 5 dB at 110 GHz and a V_π of 3.3 V at 1 MHz (3.8 V at 2 GHz), which was fed directly from a 256-GSa/s

arbitrary waveform generator (AWG, Keysight Technologies M8199B) with an analogue bandwidth of ~ 80 GHz. In the transmitter-side digital signal processing (TxDSP), the signal-generation procedures for the symbol-domain and sample-domain Volterra filters are illustrated in Fig. 2. For the sample-domain configuration used in the main experiment, the waveform was pulse-shaped and resampled to 256 GSa/s by a root-raised-cosine (RRC) filter with a roll-off factor of 0.01, followed by linear pre-equalization and sample-domain Volterra nonlinear pre-distortion. The Volterra kernel coefficients were re-estimated by ILA for each AWG swing at back-to-back, while the 80-km results used the back-to-back-trained coefficients. Different Mersenne Twister seeds were used for training and evaluation to avoid overfitting.

Polarization-division multiplexing (PDM) was emulated using a delay line and a variable optical attenuator. The transmission line consisted of a single 80-km span of pure-silica-core fibre (PSCF) with a loss of 0.152 dB/km and an effective area of $150 \mu\text{m}^2$, followed by an EDFA. Figure 3(b) shows the optical spectra of the 168-GBd signal with and without the sample-domain Volterra filter at back-to-back, as well as after the 80-km transmission. After passing through an optical bandpass filter (OBPF), the signal was detected by a coherent receiver comprising a μ ITLA local oscillator (LO) with a ~ 10 -kHz linewidth, an optical 90° hybrid, and four 90-GHz-bandwidth balanced photodetectors (BPDs). The detected signal was digitized by a 256-GSa/s digital storage oscilloscope (DSO) with an analogue bandwidth of 113 GHz. The receiver-side DSP (RxDSP) simultaneously compensated for the Tx/Rx linear responses, frequency offset, and carrier phase using a data-aided, frequency-domain 8×2 MIMO adaptive equalizer with a pilot-based digital phase-locked loop (DPPLL) [4]. The log-likelihood ratios were then calculated using a bit-metric decoder [15].

Results

Using uniform 168-GBd 64-QAM signals at the back-to-back configuration, Figure 4 shows the demodulated signal-to-noise ratio (SNR) as a function of the AWG output swing for three

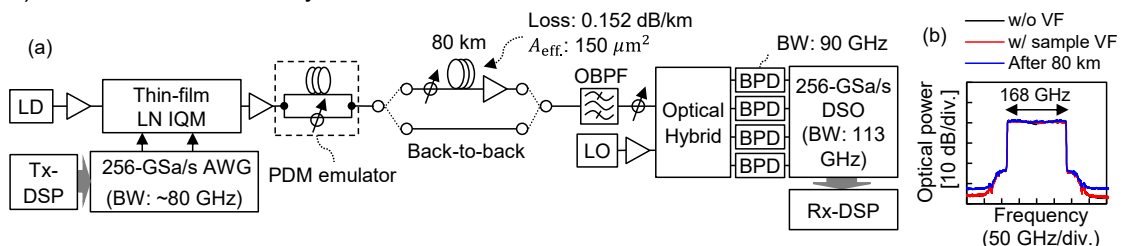


Fig. 3: (a) Experimental setup for 168-GBd PCS-32QAM signal generation and 80-km transmission. (b) Optical spectra at back-to-back (w/ and w/o sample VF) and after 80-km transmission.

transmitter DSP configurations: without a Volterra filter (w/o VF), with a symbol-domain Volterra filter (w/ symbol VF), and with a sample-domain Volterra filter (w/ sample VF). The symbol-domain Volterra filter used for comparison employed the same third-order structure, the same memory lengths of $(m_1, m_2, m_3) = (31, 31, 11)$, one sample per symbol, and the same ILA-based training procedure as the sample-domain Volterra filter. Without VF, the SNR peaked at approximately 1.1 Vpp and degraded at higher swings due to gain compression. The symbol-domain VF improved the SNR by up to 0.3 dB, while the sample-domain VF achieved improvements of 1 dB over the w/o VF and 0.7 dB at 2.5 Vpp over the symbol-domain VF configurations. Because the maximum AWG output swing of 2.7 Vpp is substantially smaller than $2V_\pi$ of the TFLN-IQM (6.6 V at 1 MHz), the TFLN-IQM operates in its near-linear regime and the dominant transmitter nonlinearity originates from the AWG's built-in variable-gain amplifier. The sample-domain VF compensates this nonlinearity more effectively than the symbol-domain VF because it directly operates on the high-PAPR, continuous-amplitude sample sequence presented to the amplifier after Nyquist shaping and linear pre-equalization.

We then optimized the PCS-QAM parameters at the same symbol rate of 168 GBd. Figure 5(a) shows the normalized generalized mutual information (NGMI) [16] and the required overall code rate as a function of the entropy for PCS-256QAM, PCS-324QAM, and PCS-400QAM. The inner LDPC code rate was increased using a family of DVB-S2 [17] LDPC codes with a puncturing method [18] until the post-LDPC BER reached the threshold of 5×10^{-5} assumed for the outer hard-decision forward error correction (HD-FEC), with an outer HD-FEC code rate of 0.9922 [19]. The resulting overall code rate, including the outer HD-FEC factor, was used to calculate the achievable bitrate and net bitrate shown in Fig. 5(b) using

$$C = \frac{2[H_{2D} - (1 - R) \log_2 M]B}{1 + P_{OH}/100}$$

where H_{2D} is the 2D-symbol entropy, R is the NGMI or code rate, M is the base QAM size before truncation [20], B is the symbol rate, and P_{OH} is the pilot overhead. The pilot overhead of 1.587% corresponds to one QPSK pilot symbol inserted per 63 data symbols. The maximum net bitrate of 2.18 Tb/s was obtained with PCS-324QAM at an entropy of 8.063 bit/2D-sym.

Finally, we evaluated the 168-GBd PCS-324QAM signal after 80-km transmission through a single PSCF span. Figure 6 shows the NGMI, code rate, achievable bitrate, and net bitrate as a

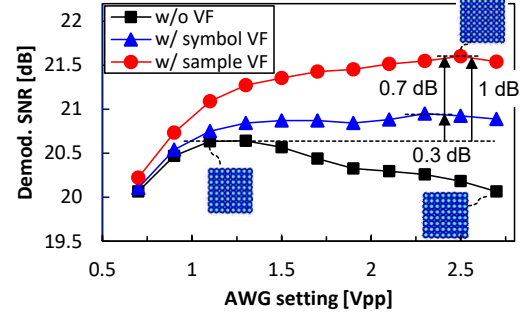


Fig. 4: Demodulated SNR vs. AWG output swing at 168-GBd uniform 64-QAM (back-to-back) for w/o VF, w/ symbol VF, and w/ sample VF.

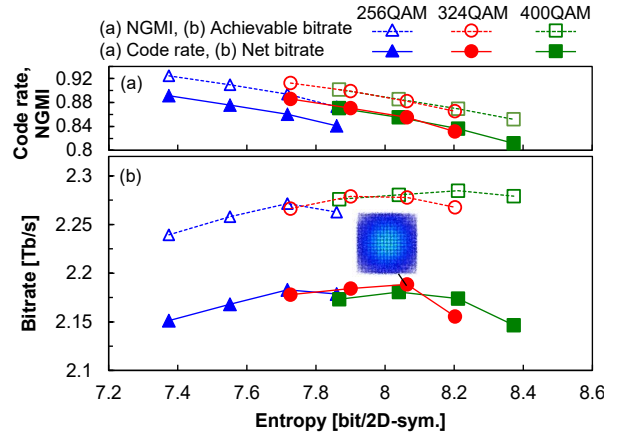


Fig. 5: PCS-QAM at 168 GBd (back-to-back). (a) NGMI and code rate vs. entropy for PCS-256/324/400QAM. (b) Achievable and net bitrate vs. entropy.

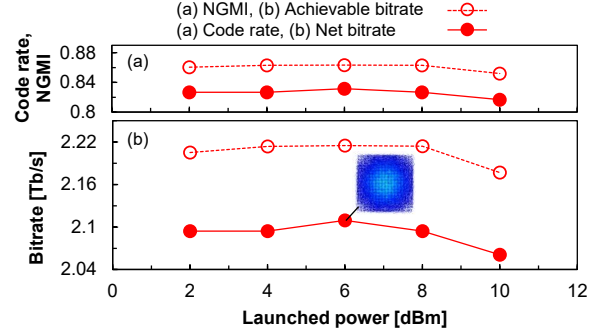


Fig. 6: NGMI, code rate, achievable and net bitrate vs. fibre-launched power for 168-GBd PCS-324QAM after 80-km transmission.

function of the fibre-launched power for an entropy of 8.063 bit/2D-sym. The optimal launched power was 6 dBm, and the net bitrate reached 2.10 Tb/s (see Fig. 1).

Conclusion

We demonstrated 168-GBd PCS-324QAM signal generation using sample-domain Volterra nonlinear pre-distortion of the high-PAPR transmitter waveform after Nyquist pulse shaping and linear pre-equalization. The sample-domain Volterra filter provided a 0.7-dB peak-SNR gain over a symbol-domain Volterra filter at 168-GBd uniform 64-QAM. Net bitrates of 2.18 Tb/s at back-to-back and 2.10 Tb/s after 80-km transmission were achieved.

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